

George Santellano, PhD

San Diego, CA | gsantellano@gmail.com | 619-204-7196

<https://gesantel.github.io/> | <https://linkedin.com/in/georgesantellano>

SUMMARY

PhD Mathematician transitioning into Machine Learning and Data Science, with hands-on experience building, evaluating, and diagnosing ML pipelines (Pytorch, Scikit-learn). Currently applies ML diagnostics to systems in production as an independent consultant – identifying data drift and model degradation. Research background in graph theory and representation theory. Five years of experience translating technical material for diverse audiences as a researcher and instructor.

TECHNICAL SKILLS

Programming: Python, SQL, MATLAB, R, C

Machine Learning: PyTorch, Scikit-learn, NumPy, Pandas, FastAI

Data & Visualization: Google Sheets, Excel, Matplotlib, Seaborn, Tableau, Google BigQuery

Version Control & Platforms: Evidently AI, Docker, Google Cloud Run, Github/Git, Linux/command line, Google Cloud, Streamlit, Hugging Face

EDUCATION

PhD in Mathematics, University of California, Riverside

Sept 2020 – Sept 2025

My dissertation developed the representation theory for matroidal Schur algebras, deriving a combinatorial formula that expresses how these complex algebraic objects decompose into structured components.

MA in Mathematics, University of Pennsylvania

Aug 2018 – May 2020

My dissertation was inspired by my summer research at the Simons Laufer MSI in Berkeley. Surveying the Polynomial Method over finite fields to longstanding problems.

BS in Applied Mathematics and Cognitive Science, University of California, San Diego

Sept 2009 – March 2014

CERTIFICATIONS

Google Data Analytics Professional Certificate

Summary Statistics in Biostatistics, Johns Hopkins University Certificate

PROFESSIONAL EXPERIENCE

ML and Data Consultant, Independent

July 2026 – Present

- **System Optimization:** Troubleshoot and resolve degraded ML system performance by evaluating serving latency, throughput, and CPU/GPU utilization.
- **Model and Data Drift Evaluation:** Diagnose performance drops using accuracy, precision, recall, f1 score, mae/rmse. Confirm data drift by comparing training vs. production sample probability distributions, KS, PSI, chi-squared test, and JS tests, plus PCA/Autoencoder reconstruction error.
- **End-to-End Data Consulting:** Architect comprehensive Scopes of Work (SOW), build data processing pipelines using Google Sheets or BigQuery, and deliver actionable business insights via Tableau and Google Slides.

AI Trainer, Handshake AI

Feb 2026 – June 2026

- Validated LLM outputs on graduate-level mathematical proofs for correctness, clarity, and rigor.
- Recognized by project managers for output, attention to detail, and breadth of subject-matter expertise.

Senior Operations Analyst, Zenefits

Oct 2014 – Jun 2016

- Automated waiver collection and processing with Python scripting, replacing two contractors and saving \$80,000 per year.
- Provided data-informed business recommendations to c-suite using SQL-based visualizations and Google Sheets.
- Partnered with engineering and product teams to translate API and data specifications into operational workflows.
- Deftly resolved customer escalations through troubleshooting and coordination with customer-facing teams.
- Managed technical relationships with external partners and maintained data integration pipelines.

TEACHING EXPERIENCE

Lecturer (Contract), UC Riverside

March 2026 – June 2026

- Designed and delivered lectures three times weekly across 3 undergraduate courses to 600+ diverse students.
- Managed 8 teaching assistants, including mediating conflict, and developing performance improvement plans.
- Wrote original final exams and managed student accommodations and concerns in collaboration with faculty..

Graduate Teaching Assistant /Associate Instructor, UC Riverside

Sept 2020 – Aug 2025

- Led discussion for undergraduate mathematics courses; communicated abstract and technical concepts clearly to diverse student populations.
- Mentored students, managed academic accommodations, and provided individualized support.
- **Winner Math Department Outstanding Teaching Award for 2024-2025.**

Remedial Math Instructor, San Francisco State University

Aug 2016 – May 2018

- Taught remedial mathematics courses, building foundational skills for students transitioning into college-level coursework.

SELECTED PROJECTS

Sobel Filter Recovery

Dec 2025

- Designed a controlled image-to-image regression study on MNIST comparing convolutional vs. fully connected networks on learning the Sobel operator of various sizes under Gaussian noise; Code written in Pytorch.
- Evaluated model behavior using MSE, accuracy metrics, and phase-diagram visualizations to characterize shifts in learned representations; published findings on Substack.

MLB Postseason Run Prediction

Sept 2025-Present

- Built an MLB StatsAPI data pipeline and trained a random forest regression model to predict team runs.
- **Predicted 87% of 2025 postseason results correctly with an average run error of ~1.4 runs per team.**
- Evaluated model using MSE.
- Containerized model as a FastAPI service with Docker, and deployed a public HTTPS endpoint on Google Cloud Run with structured request logging for latency monitoring.

Classifying National Opinion Survey Respondents by Party

Sept 2025

- Analyzed 12,000+ survey responses across 40+ demographic and political variables. Trained and compared Random Forest, KNeighbors, and AdaBoost classifiers.
- **Achieving 80% accuracy predicting party affiliation via GridSearchCV tuning, a 10% improvement over baseline.**